

A multi-scale activity-based demand generation framework for regional cities

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Abstract

This paper presents an activity-based transport demand modelling framework for regional contexts, where inter-regional commuting is important and behavioural data are often sparse. Using Greater Bendigo, a regional city in Australia, as a case study, we develop daily activity chains and subsequent activity-based demand for use in agent-based simulation models. The methodology integrates statewide travel flows with detailed local modelling through demographic synthesis, activity-chain generation, spatial location assignment, and mode-choice modelling.

The key contribution is a multi-scale location assignment framework that prioritises the mandatory activities, work and education, before allocating non-mandatory activities. Work locations are assigned to reproduce observed inter-regional commuting flows and distance distributions derived from Journey to Work (JTW) data, while education locations are allocated subject to facility-level capacity constraints. Secondary activity locations are assigned using a two-sided gravity-based approach conditioned on preceding and subsequent activity anchors, land-use-based destination attractiveness, and empirically observed mode-specific distance distributions. Trips entering and leaving the local study region are explicitly incorporated to maintain internal behavioural consistency.

The activity-based demand is assessed against observed population totals, activity timing, trip-length distributions, commuting patterns, and regional mode shares. The comparison results show close agreement with key calibration benchmarks and plausible agreement with complementary survey-based checks. The generated demand includes about 13% cross-boundary work trips, underscoring the importance of explicitly accounting for external travel in regional demand models. Overall, the proposed methodology provides a scalable and transferable approach for generating activity-based demand for regional cities while preserving cross-boundary travel consistency.

Keywords— Agent-based simulation, Synthetic population, Activity chain, Location assignment, Mandatory trips

1 Introduction

Activity-based models (ABM) rely fundamentally on the availability of realistic synthetic activity-based demand. In these models, the travel behaviour is represented through synthetic individuals characterised by sociodemographic attributes and daily activity travel patterns, commonly referred to as activity chains. By capturing decision-making at the individual level, ABMs enable detailed analysis of network interactions and policy impacts that are difficult to represent using aggregate approaches. Consequently, the construction of spatially and behaviourally consistent activity-based demand forms a critical foundation for activity-based transport modelling.

Activity-based demand generation typically involves three interrelated components: demographic synthesis, construction of daily activity patterns, and assignment of spatial locations to activities (Wang et al., 2020). Demographic synthesis establishes representative populations by matching aggregate census distributions (Rahman et al., 2010; Williamson, 2013; Hermes and Poulsen, 2012), while activity-chain generation captures the sequence, timing, and duration of activities undertaken by individuals (Duncan, 2016; Daisy et al., 2018). While demographic synthesis and activity-chain generation are relatively well-established, spatial location assignment remains a persistent challenge. Location choices directly influence trip lengths, spatial interactions, congestion levels, and mode choice outcomes, with any inaccuracies at this stage propagating throughout the simulation, leading to distorted network performance and unrealistic behavioural responses.

Conventional approaches to activity location assignment often rely on gravity-based formulations or randomised selection constrained by destination attractiveness and travel impedance (Habib, 2018; Both et al., 2021; Bhat and Koppelman, 1999). While such methods improve upon purely random allocation, they often fail to reflect critical behavioural structures, including the primacy of mandatory activities (e.g., work and education), realistic trip chaining, and interactions between local and regional travel. These limitations are especially pronounced in regional contexts, where travel demand is strongly influenced by external commuting flows and where data availability is often limited (Baqueri et al., 2019). Recent studies have sought to enhance spatial realism using observed travel behaviour, accessibility measures, or land-use information (Ziemke et al., 2019); however, few approaches explicitly integrate these improvements within a multi-scale framework that captures both regional commuting flows and fine-grained local activity assignment.

Regional ABMs, therefore, face a fundamental trade-off between spatial coverage and local realism. Models focused on small study areas often truncate external interactions or rely on limited survey samples, producing artificially localised travel patterns. This is a limitation in regional contexts, where a substantial share of mobility may involve interactions with surrounding areas, leading to an inadequate representation of the balance between local and regional travel demand. Conversely, large-scale models sacrifice neighbourhood-level detail to maintain computational feasibility. Few existing frameworks explicitly integrate coarse regional commuting flows with fine-grained local activity assignment, despite the importance of cross-boundary commuting for shaping daily mobility in regional cities.

This paper proposes a hybrid, multi-scale demand-generation framework for regional ABMs, demonstrated using the City of Greater Bendigo (CoGB), Victoria, Australia. The framework combines statewide commuting information with detailed local assignment, enabling high-level representation of the broader transport system while preserving neighbourhood-scale activity realism. Mandatory activity locations (work and education) are first assigned using Census-informed commuting flows, capacity constraints, and observed distance distributions. These locations then serve as spatial anchors for allocating non-mandatory activities using a gravity-based formulation that preserves realistic trip chaining and spatial dispersion. Cross-boundary commuting is explicitly incorporated through spatially consistent approximations, allowing external work-related demand to influence local activity patterns without requir-

ing full statewide microsimulation.

To support multimodal demand generation, the framework is complemented by a calibrated mode choice model estimated from Victorian Integrated Survey for Travel and Activity (VISTA)¹ data. Behavioural sensitivities are transferred from metropolitan estimates and combined with region-specific alternative-specific constants (ASCs), enabling the activity-based demand to reproduce observed aggregate mode shares while retaining an assumed set of behavioural responses to travel time, cost, and infrastructure attributes.

The contribution is not the individual use of census commuting flows, gravity-based assignment, or ASC recalibration, which are established techniques. Rather, the contribution lies in integrating these components into a coherent regional demand-generation workflow that preserves cross-boundary commuting while maintaining fine-grained local activity assignment. Therefore, the key contributions of this study are as follows. First, this paper introduces a hybrid multi-scale demand generation framework that integrates regional commuting flows with detailed local activity assignment for ABMs. Second, an anchor-based location assignment strategy is developed that uses mandatory activities to guide the discretionary destinations and preserve trip-chaining realism under data scarcity. Third, a pragmatic mode choice transfer approach is presented that transfers sensitivities from a larger sample and calibrates baseline preferences (ASCs) to match local mode-share targets.

The remainder of the paper is organised as follows. Section 2 reviews relevant literature on activity-based demand generation and activity location assignment. Section 3 describes the proposed methodology. Section 4 presents the results, Section 5 discusses the findings and the implications, and Section 6 concludes the paper and provides directions for future research.

2 Literature Review

2.1 Activity-based demand

Activity-based demand in an ABM represents a population of agents whose interactions and travel behaviours collectively reproduce observed mobility patterns within a study area. Generating such demand involves creating realistic populations based on demographic and socioeconomic data and serves as a core input to activity-based transport modelling (de Dios Ortúzar and Willumsen, 2024). Early ABM conceptualised travel as a consequence of activity participation and scheduling rather than independent trips (Miller, 1997; Kitamura et al., 2000), providing the theoretical foundation for contemporary ABM. According to Balmer et al. (2008), agents in ABM explicitly represent individual heterogeneity and behavioural interactions that are difficult to capture using aggregate approaches.

The generation of activity-based demand typically comprises three stages: demographic synthesis, activity-chain generation, and spatial assignment of activity locations (Wang et al., 2020). This framework has been widely adopted across different urban contexts. For instance, Sallard et al. (2020) applied this framework in São Paulo by integrating survey-based travel behaviour with spatially explicit activity assignments within a large-scale ABM. Similarly, Both et al. (2021) demonstrated its applicability at the metropolitan scale in Greater Melbourne, while large open-data implementations, such as the MATSim Open Berlin scenario, illustrate its scalability using publicly available datasets (Ziemke et al., 2019).

Methods used for demographic synthesis are commonly classified into synthetic reconstruction and reweighting approaches. Synthetic reconstruction generates baseline populations directly from census data and assigns additional attributes probabilistically, whereas reweighting adjusts the weights of survey observations to ensure consistency

¹ VISTA is a household travel survey in Victoria in which all members of surveyed households report their travel over a single specified day. <https://public.tableau.com/app/profile/vista/vizzes/#!/>

with known population distributions (Rahman et al., 2010; Hermes and Poulsen, 2012). Both techniques are well established in large-scale ABM. In particular, Williamson (2013) highlighted the effectiveness of reweighting in contexts where detailed household travel surveys are available.

Recent studies have explored the use of machine learning techniques to enhance specific components of activity-based demand generation, particularly activity participation and scheduling. For instance, Koushik et al. (2020) investigated machine learning models for refining activity-travel behaviour, while Hafezi et al. (2021) proposed a hybrid framework combining machine learning with econometric modelling. Despite their demonstrated utility, particularly in metropolitan contexts, their reliance on large volumes of high-quality data renders them less suitable for regional or data-constrained settings.

Scalability and transferability are key considerations for activity-based demand generation, especially for applications covering large geographic areas or multiple regions. Models such as TASHA (Roorda et al., 2008) and large-scale population synthesis efforts (Wang et al., 2020) highlight the importance of integrating multiple data sources and distinguishing between mandatory and secondary activities. Overall, while demographic synthesis and activity-chain generation are relatively mature components of ABM, the realism of activity-based demand remains highly sensitive to the assignment of activity locations in space.

Overall, prior research demonstrates that realistic activity-based demand in ABM arises from integrating demographic synthesis, activity scheduling, and spatial assignment of activities. Although robust methods have been developed for each component, their effectiveness depends on how they are combined and adapted to the application context. This study builds on these established approaches and adapts them to support ABM in CoGB, a regional Australian city.

2.2 Location assignment

Location assignment, or spatial assignment, determines where individual activities occur and is a core component of activity-based demand generation. By shaping travel distances, trip chaining, and the spatial distribution of demand, inaccuracies at this stage can propagate through an agent-based simulation, distorting network loading and mode shares. While demographic synthesis and activity scheduling determine who travels and when, location assignment governs how travel demand is realised in space.

Most ABM distinguish between mandatory activities (e.g., work and education) and secondary activities (e.g., shopping and leisure). Mandatory activities act as spatial anchors for daily travel patterns, whereas secondary activities are more flexible but account for a substantial share of discretionary and off-peak travel. Consistent representation of both components is therefore essential for realistic activity-based demand.

2.2.1 Mandatory location assignments

Workplace location assignment has received considerable attention due to its strong influence on commuting patterns and peak-period travel demand. In the study context, work-related travel accounts for approximately 12% of total travel time, according to the underlying VISTA data, highlighting the importance of accurately capturing workplace locations in ABM. When individual workplace data are unavailable, researchers commonly rely on aggregate employment statistics, land-use information, and observed commuting flows to assign workers to plausible job locations. Approaches based on hierarchical population synthesis, reweighting, and iterative proportional fitting have been shown to preserve consistency with observed aggregate distributions while generating individual-level assignments (Casati et al., 2015; Choupani and Mamdoohi, 2016; Agriesti et al., 2022). Simulation-based frameworks

further improve realism by incorporating individual travel behaviour and network conditions into the assignment process (Farooq et al., 2013). Despite these advances, many workplace assignment methods rely on strong assumptions regarding spatial structure and data availability.

Education location assignment has also been extensively explored, reflecting the importance of school and university travel in daily activity patterns. Hierarchical and probabilistic models have been used to capture heterogeneity in education choice behaviour (Sun et al., 2018), while large-scale applications demonstrate scalability using census and travel diary data in metropolitan contexts (Wickramasinghe et al., 2020). Empirical evidence also suggests that proximity-based assumptions may not fully reflect observed travel-to-school behaviour, with implications for travel demand and residential location choice (Liao and Dai, 2022).

While these approaches perform well in metropolitan regions, they are less directly transferable to regional settings, where employment and education catchments frequently extend beyond administrative boundaries and a substantial share of mandatory travel involves external commuting. Addressing such boundary effects while maintaining internally consistent location assignments motivates the approach adopted in this study.

2.2.2 Secondary location assignments

Secondary location assignments are essential for capturing discretionary and non-mandatory travel demand, including shopping, leisure, and recreational trips, which together account for approximately 41% of total travel time in the underlying VISTA data for the case study. Although these activities are more flexible than mandatory ones, they play a central role in shaping off-peak travel patterns, network performance, and overall urban dynamics. Accurate representation of secondary activities is therefore critical for achieving realistic activity-based demand.

Most studies assign secondary activity locations using gravity-based or probabilistic approaches that balance distance decay with destination attractiveness. For example, Hörl and Axhausen (2023) proposed a relaxation-discretisation framework to refine non-work location assignments under spatial constraints, while Hörl and Balac (2021) demonstrated the use of open-source data to model secondary activities at scale. Empirical evidence consistently identifies travel distance and destination attractiveness as dominant determinants of secondary destination choice (Huang and Levinson, 2015; Simini et al., 2012). More recent hybrid approaches combine traditional travel surveys with emerging data sources, such as mobile phone data, to improve representation in data-scarce contexts (Bwambale et al., 2021; Agriesti et al., 2022). Applications that integrate both mandatory and secondary activities further demonstrate that probabilistic and hop-based methods can reproduce observed spatial patterns and travel distances (Both et al., 2021), highlighting substantial progress in secondary location assignment methodologies.

In regional contexts, secondary activities are frequently influenced by travel beyond the study area, including discretionary trips associated with employment, education, and services located outside regional boundaries. Capturing these interactions requires secondary location assignment methods that remain consistent with local spatial structure while accounting for external and statewide travel, a challenge directly addressed in this study.

Overall, the literature demonstrates that realistic activity-based demand in ABM requires the coherent integration of demographic synthesis, activity-chain generation, and the spatial assignment of activity locations. While well-established methods exist for each component, most applications have been developed for large metropolitan regions and often assume closed or weakly interacting system boundaries. Location assignment approaches for both mandatory and secondary activities have advanced substantially, leveraging aggregate data, probabilistic models, and emerging data sources to reproduce observed spatial patterns and travel distances. However, in regional settings characterised by strong cross-boundary commuting and discretionary travel, these assumptions can limit behavioural realism. Building on established ABM frameworks, this study adapts and integrates activity-based demand gener-

ation methods to explicitly account for regional boundary effects while preserving internally consistent spatial and behavioural structures.

3 Methods

This section describes the methodological framework used to generate activity-based demand for CoGB, integrating multiple datasets and modelling components, as summarised in Figure 1. The overall process comprises four stages: data preparation, activity-chain generation, spatial assignment of activity locations, and mode-choice modelling.

Different subsets of the available data were used for different components of the framework. Weekday VISTA records for CoGB were used to derive observed activity sequences, participation patterns, and time-of-day distributions for activity-chain generation and behavioural validation. The Victorian Integrated Transport Model (VITM2) synthetic population for the 2018 base year was used as the demographic base population, providing individual-level representations of residents and workers with attributes aligned to 2016 Census distributions at the Statistical Area Level 2 (SA2)² level and home locations assigned to plausible residential street addresses. In addition, Australian Bureau of Statistics (ABS) Journey to Work (JTW)³ data were used to represent regional commuting flows, and a larger Greater Melbourne⁴ VISTA sample was used to estimate transferable mode choice sensitivities where local revealed-preference data in CoGB were insufficient. This separation of roles across datasets allowed the framework to preserve local realism where possible while drawing on broader statewide information where necessary.

3.1 Data preparation and study area

3.1.1 Statewide network generation

The modelled study area included the CoGB (represented by SA3 region 20201) along with its two adjacent SA3 regions 20202 and 20203, along with a 10-kilometre buffer surrounding the CoGB for the road network to reduce edge effects in network-based analyses (Figure 2). This enabled a more accurate representation of through-traffic and ensured continuity across road segments spanning administrative boundaries. To capture regional connectivity without burdening network analysis with a detailed network for areas outside the CoGB and its buffer, highways (but not minor roads) for the whole of the State of Victoria were also incorporated into the network. The base road network was derived from Open Street Map (OSM), which provides detailed information on streets, intersections, and pathways. Public Transport (PT) infrastructure was added using General Transit Feed Specification (GTFS) data from Public Transport Victoria (PTV).

3.1.2 Data preparation

Activity-based demand generation drew on three primary data sources: the 2016 ABS JTW dataset, the Victorian Integrated Transport Model (VITM) synthetic population for the 2018 base year, and VISTA travel survey records. The VISTA data were used in two ways: the CoGB sample was used to derive local activity participation, sequencing, and timing patterns, while a larger Greater Melbourne sample was used for mode choice model estimation, where the regional sample size was insufficient for robust parameter estimation. The generated demand should therefore be interpreted as an approximate 2018 baseline demand. The 2018 VITM synthetic population provides the demographic

²SA2 units, defined by the Australian Bureau of Statistics, typically represent suburbs with approximately 10,000 residents.

³<https://www.abs.gov.au/articles/australias-journey-work>

⁴Melbourne is the capital city of the state of Victoria, Australia. The City of Greater Bendigo is also located in Victoria.

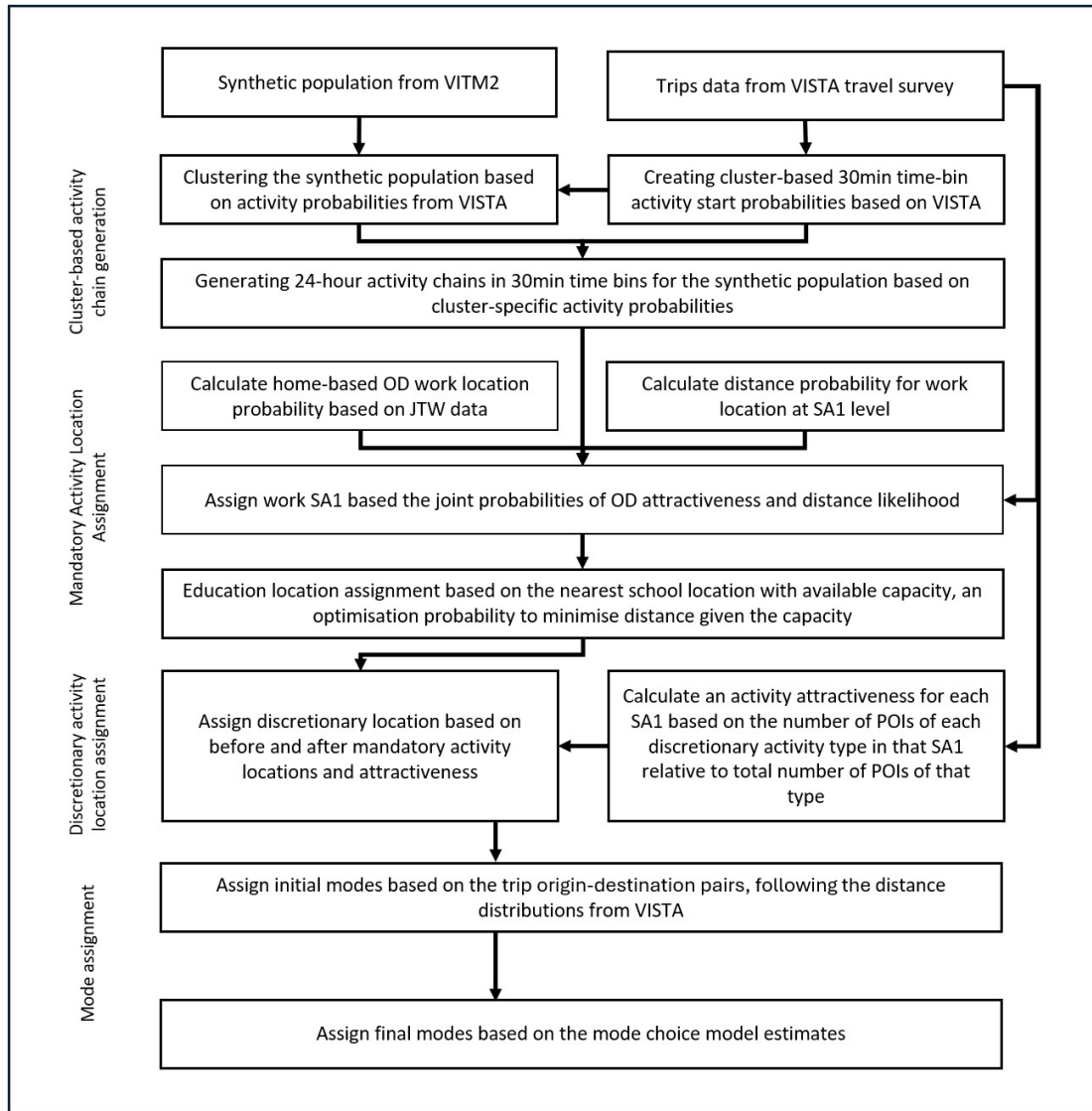


Figure 1: Overview of the activity-based demand generation methodology

base, the 2016 JTW data provide the most detailed available commuting structure, and pooled VISTA records are used to obtain stable behavioural distributions in a regional context where local survey samples are limited. This requires the assumption that the main commuting, activity timing, and mode choice structures were sufficiently stable across these data periods for the purpose of baseline demand generation. We use CoGB VISTA (2012–2023) for activity timing/validation and Greater Melbourne VISTA (2012–2020) for mode-choice estimation.

The JTW data were used to quantify commuting flows into the CoGB from external SA3 regions. For each contributing SA3, the proportion of workers travelling to CoGB was calculated and applied to the VITM population to selectively incorporate external full-time workers, scaled by SA3-specific workforce shares. All individuals included in the study, including inbound commuters, were assigned full activity chains to ensure a consistent behavioural

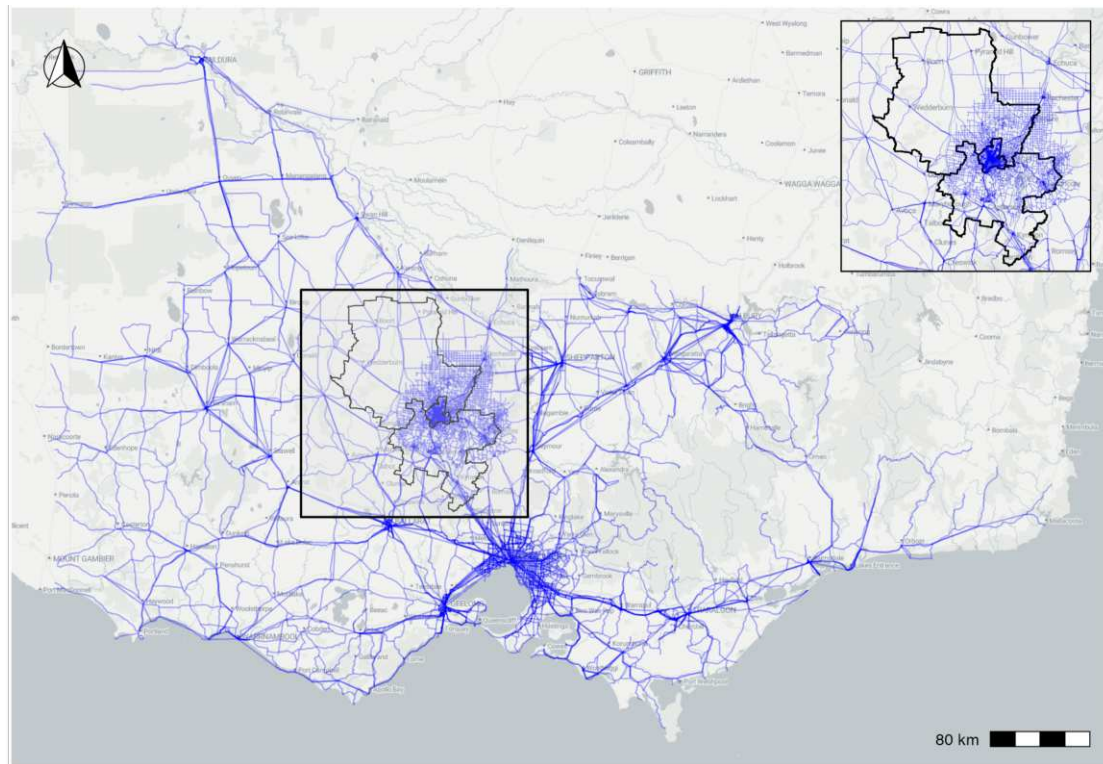


Figure 2: Statewide highways with detailed CoGB network

representation.

This process yielded a synthetic population of 164,129 individuals, of whom 158,506 reside within CoGB. As an internal consistency check against ABS workforce statistics, adjusted for population growth between 2016 and 2018, the procedure produced 5,774 modelled inbound work trips compared to an expected 5,825, corresponding to a deviation of only 0.9%.

To support fine-grained spatial assignment, individuals from the VITM population, originally defined at the SA2 level, were disaggregated from SA2 to SA1 using ABS joint age–gender distributions. The assignment was performed via stratified probabilistic sampling, ensuring consistency with the observed SA1-level demographic structure while preserving aggregate SA2 characteristics.

Daily activity sequences and trip attributes were obtained from VISTA. Main activity labels and trip purposes were consolidated into a standardised set to ensure consistency with VITM population attributes, and to facilitate activity-chain generation and subsequent spatial location assignment. Activity purposes were harmonised into the categories *Home*, *Work*, *Study*, *Shop*, *Personal*, *Social/Recreational*, *Pickup/Dropoff/Deliver*, and *Other*, while employment and education statuses were mapped to simplified main activity tags (Table 1).

⁵TAFE refers to Technical and Further Education institutions in Australia, which provide vocational education and training.

Table 1: Main Activity Category Mapping from VISTA and VITM2 to Simplified Main Activity Tags

Original Main Activity	Main Activity Category
Casual Work, Part-time Work, Part-time worker	Part-time worker
Full-time Work, Full-time worker	Full-time worker
Retired	Retired
Not yet at School, Not in school (preschool)	Preschool
Primary School	Primary school
Secondary School	Secondary school
Full-time TAFE ⁵ /Uni, Part-time TAFE/Uni, Tertiary education and part-time worker, Tertiary education (non-worker), Tertiary education and full-time worker	Tertiary education
Keeping House, Unemployed, Volunteer, Other Education, Other	Other

3.1.3 Clustering analysis

To capture heterogeneity in daily travel behaviour, individuals were grouped into distinct behavioural classes using a data-driven clustering approach. Weekday trip records from VISTA were used to compute, for each individual, the probability of participating in key activity types, including work, study, shopping, personal, and social or recreational activities. These activity participation probability vectors served as the input features for clustering.

Hierarchical clustering was performed using Ward’s minimum variance method (Ward Jr, 1963), which promotes compact and well-separated clusters. The optimal number of clusters was determined using the gap statistic, resulting in five distinct behavioural groups. These groups were identified empirically from activity participation patterns and were then interpreted using the dominant main-activity profiles observed within each cluster, as summarised in Table 2. The labels shown in the table therefore represent an interpretation of the empirical clusters rather than ex ante population segments.

Following clustering, the VISTA trip records were partitioned into cluster-specific subsets. This enabled the derivation of cluster-specific time-of-day activity distributions, allowing activity timing patterns to vary systematically across behavioural groups. These distributions were subsequently used to generate daily activity schedules, ensuring that temporal structure reflects observed heterogeneity in activity participation across the population.

3.2 Creating the activity chains

Activity chains describe the ordered sequence, timing, and duration of activities undertaken by individuals over a day and form the behavioural core of ABM. Unlike trip-based approaches, ABMs represent travel as resulting from activities that are temporally ordered and interdependent, enabling a coherent 24-hour representation of travel demand.

In this study, weekday activity chains are generated using observed travel diaries from the VISTA. A probabilistic distribution-matching approach is adopted, closely following Both et al. (2021), with the objective of reproducing aggregate activity participation rates and time-of-day distributions without replicating individual survey records.

Table 2: Clustered main activity categories derived from hierarchical analysis

Cluster	Main activity category
1	Full-time worker, Part-time worker
2	Other, Retired
3	Tertiary education
4	Primary school, Secondary school
5	Preschool

3.2.1 Derivation of empirical activity distributions

Activity episodes are reconstructed from VISTA trip records by interpreting the interval between consecutive trips as the duration of the activity performed at the destination of the previous trip. The first activity of the day is assumed to start at midnight, and the final activity ends at the end of the day, yielding complete daily schedules.

Detailed activity labels were aggregated into the categories: *Home, Work, Study, Shop, Personal, Social/Recreational, and Pickup/Dropoff/Deliver*. The day was discretised into 48 half-hour time bins, and weighted empirical distributions of activity start times and end times were computed for each activity type. Survey expansion weekday weights ensure population-representative distributions.

To capture behavioural heterogeneity, these distributions were estimated separately for each behavioural cluster identified as per Table 2, allowing activity participation and timing patterns to vary systematically across population groups.

3.2.2 Probabilistic generation of daily activity chains

Daily activity chains were synthesised sequentially from the start to the end of the day. At each time bin, activity types that are under-represented relative to empirical targets were given higher selection probabilities. This corresponds to an adaptive distribution-matching mechanism that progressively aligns the generated distributions with observed targets. For activity type a at time bin t , the adaptive selection weight is defined as

$$\Delta_{a,t} = \max \left(0, D_{a,t}^s - D_{a,t}^{s'} \right), \quad (1)$$

where $D_{a,t}^s$ is the target start-time distribution derived from VISTA and $D_{a,t}^{s'}$ is the distribution achieved so far in the activity-based demand. This mechanism ensures convergence toward observed aggregate timing patterns.

Once an activity is selected, its end time is sampled from the corresponding conditional distribution, and the algorithm then advances to the next time bin. Consecutive activities of the same type are merged to maintain behavioural plausibility. In rare cases where no activity is generated, a single all-day *Home* activity is assigned.

Each synthetic individual receives an activity chain drawn from cluster-specific distributions, preserving heterogeneity while remaining computationally scalable. These temporally consistent activity schedules form the behavioural input to the subsequent spatial location assignment procedure.

3.3 Location assignment

This section describes the spatial assignment of activity locations across the study area. Given temporally consistent daily schedules for each agent, locations were assigned sequentially using a hierarchical procedure: residential

locations were assigned first, followed by mandatory activities (work and education), and finally non-mandatory activities. This structured approach ensured consistency between individual-level behaviour and observed aggregate travel patterns (Tiwari et al., 2025a,c,b).

The study area comprised SA3 region code 20201, which defines the urban boundary of the CoGB, along with region code 20202 and 20203, which together comprise the CoGBs SA4 region. A comprehensive database of valid street addresses was compiled (Department of Transport and Planning, 2026) and classified into five destination categories: *Home*, *Work*, *Education*, *Commercial*, and *Park*. These categories define destination availability and constrain feasible spatial choices for different activity purposes. As shown in Figure 3, residential addresses are broadly distributed across suburban areas, whereas employment and commercial locations are concentrated within the central business district. This spatial structure provided the foundation for the subsequent assignment of mandatory and non-mandatory activities.

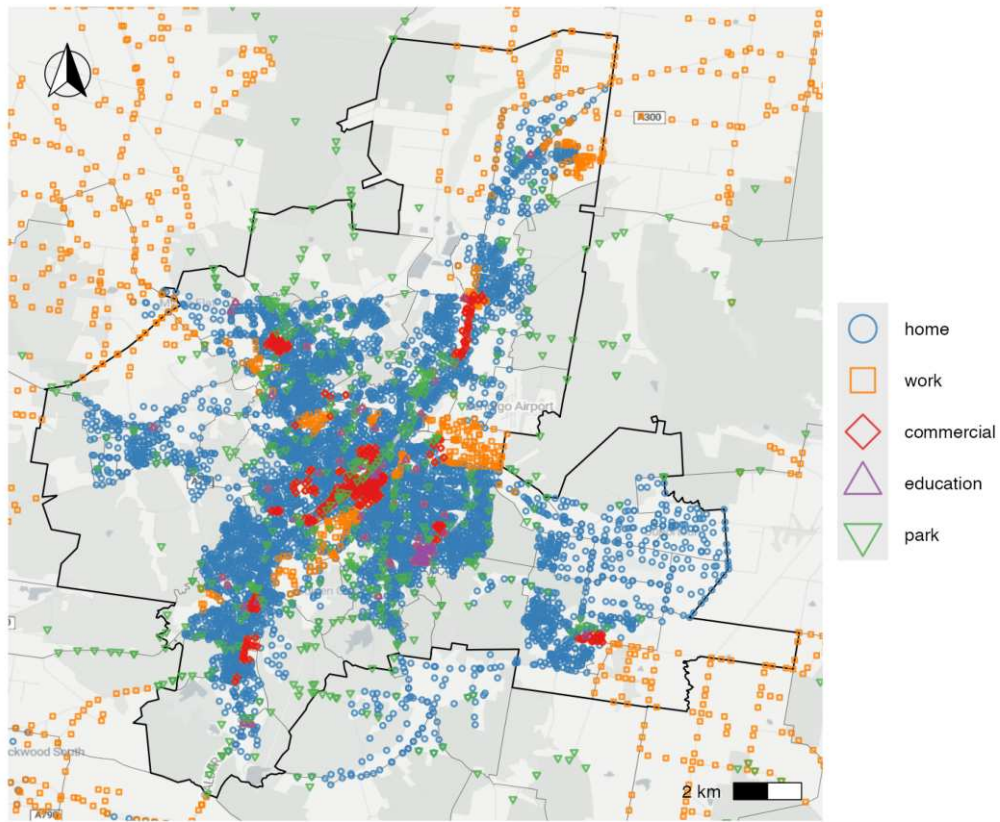


Figure 3: Available addresses within CoGB

3.3.1 Mandatory locations

Mandatory activities, namely work and education, were assigned using activity-specific strategies that reflect distinct behavioural and institutional constraints. In both cases, assignment was formulated as a constrained probabilistic process that balances travel impedance and destination capacity while reproducing observed aggregate patterns.

Work Location Assignment: This assignment matches SA3 OD flows and SA3-conditional distance distributions, and it also aligns with the statewide commuting distance profile. Work locations were assigned at the SA1 level using commuting flows from the ABS 2016 JTW data. The assignment was designed to satisfy three empirical constraints jointly: (i) inter-regional commuting flows between SA3 regions, (ii) SA3-conditional commuting distance distributions, and (iii) the global commuting distance distribution.

Let $i \in \mathcal{W}$ denote an employed individual in home SA1 h_i within home SA3 c . From the JTW data, commuting flow probabilities $P(r | c)$, where r denotes the destination's work SA3. Workers were first allocated to destination SA3s according to these probabilities. For each home SA3 c , the number of workers assigned to destination SA3 r was computed as

$$N_{c,r} = \text{round}(N_c \cdot P(r | c)), \quad (2)$$

where N_c is the total number of employed residents in SA3 c . A constrained rounding procedure preserved marginal totals across destination regions.

Conditional on the assigned work SA3 r , candidate workplace locations were restricted to SA1 regions within r . Each candidate SA1 j was associated with an expected employment attractiveness weight $P_{\text{loc,expected},j}$, derived from SA1-level employment capacity data, and a network-based commuting distance d_{ij} from the worker's home SA1 h_i .

To reproduce observed commuting distances, two complementary likelihood components were incorporated. A regional distance likelihood $P_{\text{reg},j} = P(d_{ij} | c, r)$ captured home-work SA3 distance patterns, while the global distance likelihood $P_{\text{glob},j} = P(d_{ij})$ enforced consistency with the overall statewide commuting profile. Both components were updated dynamically using an IPF mechanism, allowing subsequent assignments to correct emerging deviations.

Empirical commuting distance distributions derived from the ABS JTW data exhibited a pronounced peak at intermediate distances (approximately 2.5–3.5 km), followed by a monotonic decline for longer trips (Figure 4). While the overall shape is broadly consistent with the log-normal form commonly observed in travel distance distributions, the empirical distributions were used directly by discretising them into distance bins to define both the regional distance likelihood $P(d_{ij} | c, r)$ and the global distance likelihood $P(d_{ij})$, ensuring that workplace assignments reproduced observed distance-decay behaviour rather than relying on assumed functional forms.

To prevent over-allocation of highly attractive work locations, an adaptive location prior was introduced based on the discrepancy between expected and realised workplace allocations:

$$P_{\text{loc},j} = \frac{\max(0, P_{\text{loc,expected},j} - P_{\text{loc,actual},j})}{\sum_k \max(0, P_{\text{loc,expected},k} - P_{\text{loc,actual},k})}, \quad (3)$$

where $P_{\text{loc,actual},j}$ denotes the cumulative share of workers already assigned to SA1 j . If no imbalance remained, a uniform prior over candidate locations is applied.

For each worker i and candidate workplace SA1 j , an unnormalised assignment score was then computed as

$$S_{ij} = \frac{P_{\text{loc},j} + \alpha P_{\text{reg},j} + \beta P_{\text{glob},j}}{\eta(d_{ij})}, \quad (4)$$

with $\alpha = \beta = 2$ in this study. The denominator $\eta(d_{ij})$ introduced a soft distance moderation term,

$$\eta(d_{ij}) = 1 + \frac{d_{ij}}{d_{95}}, \quad (5)$$

where d_{95} denotes the 95th percentile of observed commuting distances. The final assignment probability is obtained by normalising these scores across all candidate SA1 locations in the assigned destination SA3:

$$P_{ij} = \frac{S_{ij}}{\sum_{k \in \mathcal{J}_i} S_{ik}}. \quad (6)$$

The values of α and β were selected to give comparable influence to the regional and global distance-matching terms while retaining a role for local employment attractiveness. They are transparent balancing parameters used to preserve agreement with observed commuting distributions while avoiding overfitting in a data-sparse regional context.

Workers were processed sequentially, with SA1 employment totals, SA3 commuting flows, and distance distributions updated after each assignment. This dynamic balancing steered subsequent allocations toward underrepresented spatial and distance targets while preserving stochastic variability, yielding workplace locations consistent with observed commuting patterns at multiple spatial scales.

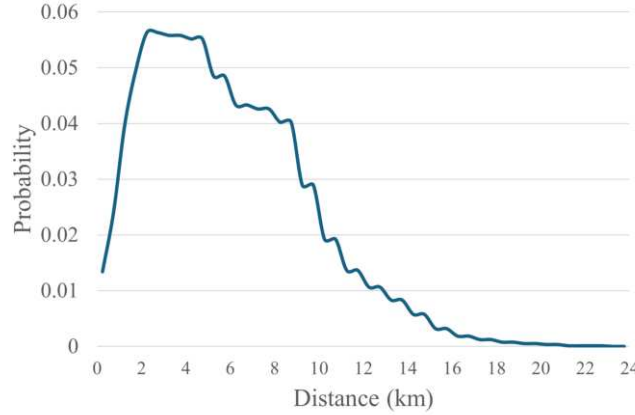


Figure 4: Distance-based probability for choosing a work destination.

Education location assignment: Education locations were assigned to ensure spatially plausible enrolment patterns while strictly respecting level-specific capacity constraints. Students were classified into three mutually exclusive groups: primary (U_p), secondary (U_s), and tertiary (U_t), based on their recorded main activity, with $U = U_p \cup U_s \cup U_t$ denoting the full student population. Each education facility $s_j \in F = \{s_1, \dots, s_m\}$ was characterised by geographic location and level-specific capacities $C_p(s_j)$, $C_s(s_j)$, and $C_t(s_j)$, obtained from the Australian Curriculum, Assessment and Reporting Authority (ACARA)⁶ and Victorian Government education statistics⁷.

Let $u_i \in U$ denote a student residing in SA1 h_i , represented by its centroid, and let $d(i, j)$ denote the network-based distance between h_i and school s_j .

Primary students ($u_i \in U_p$) were assigned deterministically to the nearest feasible school, reflecting strongly localised enrolment behaviour:

$$s_i^* = \arg \min_{s_j \in S} d(i, j) \quad \text{s.t.} \quad C_p(s_j) > 0, \quad (7)$$

after which the corresponding capacity was decremented. If the nearest school was at capacity, the next closest feasible school was selected iteratively.

Secondary and tertiary students ($u_i \in U_s \cup U_t$) were assigned using a probabilistic distance-based approach subject to remaining capacity constraints. For each student i , candidate schools were restricted to those with available

⁶<https://www.acara.edu.au/contact-us/acara-data-access>

⁷<https://www.vic.gov.au/statistics-victorian-schools-and-teaching>

capacity, forming the feasible set $\mathcal{S}_i$. Assignment probabilities are defined as:

$$P(i, j) = \frac{1 / \max(\varepsilon, d(i, j))}{\sum_{k \in \mathcal{S}_i} 1 / \max(\varepsilon, d(i, k))}, \quad (8)$$

favouring geographically proximate institutions while preserving stochastic variability. School capacities were updated immediately following each assignment.

To improve spatial efficiency while retaining stochastic variability, the initial assignment was followed by a local-improvement procedure that sought to reduce total student travel distance subject to capacity constraints. The aggregate system distance was defined as

$$D = \sum_{i \in U_s \cup U_t} d(i, s_i), \quad (9)$$

where s_i denotes the school currently assigned to student i . Iterative reassignment was permitted when an alternative feasible school yielded a strict reduction in D without violating capacity constraints. This refinement continued until no further distance-reducing reassignment was possible, ensuring a locally distance-efficient allocation.

Any students remaining unassigned due to capacity exhaustion were finally allocated to the nearest school with residual capacity, ensuring a complete and feasible assignment. Overall, the approach combined deterministic nearest-neighbour allocation for primary education with probabilistic, capacity-constrained distance-based assignment and local distance minimisation for secondary and tertiary education, yielding realistic enrolment patterns while maintaining spatial and institutional consistency.

This formulation should be interpreted as a pragmatic spatial allocation procedure for transport demand generation rather than a full model of school choice. In particular, the nearest-feasible assignment for primary students reflects the strongly local nature of many primary school trips in regional settings, but does not explicitly represent school catchment policies, private school selection, household preferences, or parental decision-making. However, it does not explicitly represent factors such as private school selection or parental preferences. The approach is intended to provide spatially plausible education locations under available data constraints while preserving network-relevant travel patterns.

3.3.2 Non-mandatory locations (secondary activities)

The secondary location assignment method is the main extension beyond a standard gravity model. It selects secondary destinations to fit between the preceding and subsequent fixed activities in the daily schedule. Secondary activities such as shopping, personal business, and social or recreational trips were assigned using a gravity-based framework that extends the approach of [Both et al. \(2021\)](#) by explicitly conditioning destination choice on both preceding and subsequent mandatory activities. Home, work, and education locations act as spatial *anchors*, ensuring that discretionary trips are embedded within coherent daily trajectories rather than being allocated independently of the surrounding activity context.

In practical terms, the aim was to assign each secondary activity to a location that is plausible from both sides of the activity chain. A destination should be consistent with the observed travel distances for the chosen mode from the preceding anchor, yet allow the individual to continue toward the subsequent anchor without creating unrealistic detours. This is particularly important in regional settings, where the smaller number of major destinations can otherwise lead to spatially scattered or behaviourally implausible daily schedules.

Let $\mathcal{A}_i = \{a_0, a_1, \dots, a_L\}$ denote the ordered set of mandatory anchors for individual i , where each anchor a_ℓ corresponds to a known SA1 location. Secondary activities occurring between two consecutive anchors a_u and a_v were assigned sequentially. The objective was to allocate each activity to a plausible SA1 such that (i) travel

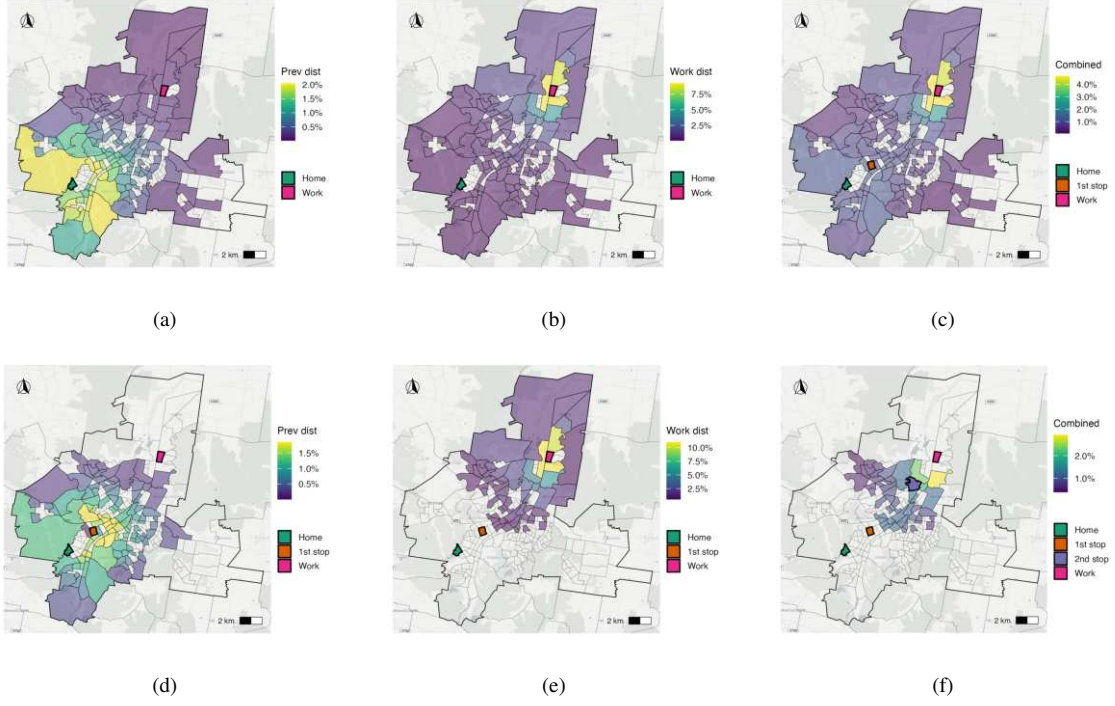


Figure 5: Selecting non-mandatory locations for a cycling trip from home to work, showing the distance probability from home (a), to work (b), and combined probability (c) for the first stop, along with distance probability from the first stop (d), to work (e), and combined probability for the second stop.

distances remain consistent with empirically observed, mode-specific trip-length distributions, (ii) destinations reflect land-use-based attractiveness and opportunity availability, and (iii) activities remain spatially constrained between neighbouring anchors in the daily schedule.

For each secondary activity, a provisional arriving travel mode $m \in \{\text{walk, bike, pt, car}\}$ was first sampled from the local mode distribution at the preceding anchor h_u , following the same procedure as in Both et al. (2021). This provisional mode was used only as a reachability and distance-feasibility constraint during location assignment. Final mode assignment was undertaken later using the MNL mode choice model. Therefore, the sampled mode in this step should be interpreted as a spatial feasibility proxy rather than as the final behavioural mode choice.

Candidate destination SA1s are then restricted using a reachability-based corridor constraint. Let $d(s_j, h)$ denote the network distance between candidate SA1 s_j and anchor h , and let $\hat{d}_m^{95}(h)$ be the 95th percentile of observed trip lengths for mode m from origin h , estimated from VISTA. The number of travel “jumps” required to return from s_j to the subsequent anchor h_v was approximated as

$$J_m(s_j, h_v) = \left\lceil \frac{d(s_j, h_v)}{\hat{d}_m^{95}(h_v)} \right\rceil. \quad (10)$$

Let K denote the number of remaining activities before the next anchor. A candidate SA1 was considered admissible if $J_m(s_j, h_v) \leq K$ and if it contained at least one destination of the relevant activity category (e.g., commercial locations for shopping or parks for recreational activities). This corridor-based filtering step ensured that

secondary activities remain spatially feasible within the individual’s daily schedule and prevents implausible detours that would require excessive backtracking to reach subsequent mandatory activities.

Within the resulting admissible set, destination choice probabilities were computed using an anchor-based gravity formulation. For each candidate SA1, anchor-specific desirability scores were evaluated separately with respect to both the preceding anchor h_u and the subsequent anchor h_v . Each score combined four conceptually distinct components. First, a local distance likelihood captured the plausibility of the anchor-to-destination distance under mode-specific trip-length distributions estimated from VISTA, improving consistency with observed trip-length patterns. Second, a dynamic distance-correction component accounted for deviations between the realised and empirical distance distributions as assignments progressed, allowing the model to progressively compensate for emerging imbalances. Third, a local destination attractiveness term reflected the density of relevant land-use opportunities within each SA1, favouring locations with greater availability of suitable destinations. Finally, a regional correction component aligned realised destination shares with expected SA3-level patterns, preventing systematic over- or under-allocation to specific subregions.

These components were combined to produce anchor-specific gravity scores, which were normalised to yield anchor-specific destination-choice probabilities. To ensure spatial coherence of activity chains, the final destination probability blended the distributions associated with both neighbouring anchors:

$$P_i(s_j | a_u, a_v, m, c) = \lambda \pi_{h_u}(s_j) + (1 - \lambda) \pi_{h_v}(s_j), \quad (11)$$

where π_{h_u} and π_{h_v} denote the normalised gravity probabilities relative to the preceding and subsequent anchors, respectively, and $\lambda = 0.5$ in this study. This two-sided formulation favoured destinations that are jointly plausible from both sides of the activity sequence, thereby reinforcing spatial continuity and realistic trip chaining. An equal weighting was adopted as a neutral specification that gives the preceding and subsequent anchors the same influence in the absence of evidence favouring one side of the chain over the other.

The destination SA1 for each secondary activity was sampled from this blended distribution. If no admissible candidate exists, the activity was conservatively assigned to the preceding anchor location. Secondary activities were processed sequentially along each individual’s daily schedule. After each assignment, realised distance histograms and regional destination counters are updated, allowing subsequent allocations to progressively correct emerging deviations while preserving individual-level stochasticity.

Overall, the proposed two-sided, anchor-based gravity framework extended [Both et al. \(2021\)](#) by explicitly embedding secondary activity locations within spatial corridors defined by neighbouring mandatory activities, while dynamically enforcing empirical distance distributions and regional destination patterns. This approach yielded spatially coherent daily trajectories and behaviourally realistic activity-based demand suitable for regional-scale agent-based transport simulation. The complete location assignment procedure is summarised in Algorithm 1.

3.4 Mode assignment

3.4.1 Mode choice model

Due to the limited availability of revealed-preference data in CoGB, mode-choice parameters were estimated using a larger VISTA sample from Greater Melbourne and then transferred to the CoGB model. This transferability assumption was based on the premise that the underlying behavioural responses to key mode choice attributes, including travel time, monetary cost, and cycling stress, were sufficiently similar across the two contexts, while observed differences in mode shares arose primarily through differences in urban form, transport supply, and resulting level of

Algorithm 1: Hierarchical location assignment procedure

Input : Agents with fixed home SA1 h_i (home SA3 c_i) and daily schedules;
Work: JTW flows $P(r \mid c)$, distance targets $P(d \mid c, r)$ and $P(d)$, SA1 employment weights $P_{\text{loc,expected}}$, distances $d(\cdot, \cdot)$, parameters α, β ;
Education: School set $\mathcal{F}$ with capacities C_p, C_s, C_t , distances $d(i, j)$, ε , local improvement limit I_{max} ;
Secondary: SA1 set $\mathcal{R}$, category attractiveness $a_c(\cdot)$, mode-specific distance distributions, counters $\hat{p}_m, \hat{q}_c$, weights $\delta_1 \dots \delta_4$, blending parameter λ .
Output: Assigned workplace w_i , education facility s_i , and SA1 locations for all secondary activities (with initial mode).

Step 1: Work locations

- 1 Compute $N_{c,r} \leftarrow \text{round}(N_c P(r \mid c))$ (marginals preserved)
- 2 Assign each worker i a destination SA3 r_i
- 3 Initialise employment and distance counters
- 4 **foreach** worker i **do**
 - 5 Construct candidate SA1 set $\mathcal{J}_i = \{j \in \mathcal{R} : j \in r_i\}$ and distances d_{ij}
 - 6 Compute adaptive location prior $P_{\text{loc},j}$ and distance likelihoods $P_{\text{reg},j}, P_{\text{glob},j}$
 - 7 Compute scores $P_{ij} \propto (P_{\text{loc},j} + \alpha P_{\text{reg},j} + \beta P_{\text{glob},j}) / \eta(d_{ij})$
 - 8 Sample workplace w_i
 - 9 Update SA1 employment totals and distance counters

Step 2: Education locations

- 10 Initialise remaining capacities $\tilde{C}_p, \tilde{C}_s, \tilde{C}_t$
- 11 **foreach** primary student $u_i \in U_p$ **do**
 - 12 Assign nearest feasible school s_i ; decrement $\tilde{C}_p(s_i)$
- 13 **foreach** student $u_i \in U_s \cup U_t$ **do**
 - 14 Form feasible set $\mathcal{F}_i$
 - 15 **if** $\mathcal{F}_i \neq \emptyset$ **then**
 - 16 Sample s_i using inverse-distance weights; decrement capacity
- 17 Assign any unallocated students to the nearest school with residual capacity
- 18 Optionally apply capacity-feasible local improvements for at most I_{max} iterations

Step 3: Secondary activities

- 19 **foreach** agent i **do**
 - 20 Extract ordered anchor set $\{a_0, \dots, a_L\}$ with fixed SA1 locations
 - 21 Initialise remaining activities count K
 - 22 **for** $\ell = 0$ **to** $L - 1$ **do**
 - 23 Set anchor SA1s (h_u, h_v) corresponding to a_ℓ and $a_{\ell+1}$, and identify secondary activities between them
 - 24 **foreach** secondary activity p **do**
 - 25 Sample arriving mode $m \sim P(\cdot \mid h_u)$
 - 26 Build corridor candidate set $\mathcal{S}_i^{(u,v)}$ using reachability (J_m) and category validity
 - 27 **if** $\mathcal{S}_i^{(u,v)} = \emptyset$ **then**
 - 28 Assign SA1(p) $\leftarrow h_u$; decrement K ; **continue**
 - 29 Compute anchor-specific gravity distributions $\pi_{h_u}(\cdot)$ and $\pi_{h_v}(\cdot)$
 - 30 Sample $s^* \sim \lambda \pi_{h_u} + (1 - \lambda) \pi_{h_v}$; Assign SA1(p) $\leftarrow s^*$, update $\hat{p}_m, \hat{q}_c$, and set $h_u \leftarrow s^*$
 - 31 Decrement K

service. In this sense, the model assumes that the same broad behavioural mechanisms operate in both settings, but that they are expressed differently because travellers face different spatial structures, travel times, costs, and network conditions.

Mode choice behaviour was modelled using a Multinomial Logit (MNL) framework, with walking, cycling, driving, and PT as alternatives. The estimation dataset comprised 66,223 weekday trips from Greater Melbourne VISTA (2012–2020) with valid travel-time information available for all four modes. The resulting coefficients for travel time, monetary cost, and cycling stress were directly transferred to the CoGB model. To improve consistency with local observed mode shares, the ASCs were subsequently recalibrated for CoGB while holding the marginal utility parameters fixed. This approach allowed the model to retain behaviourally interpretable trade-offs estimated from the larger metropolitan sample while adapting baseline mode preferences to the regional context.

Travel times for car, walk, and PT were estimated using the R5 routing engine via the `r5r` package in R, combining OSM and GTFS data (Pereira et al., 2021). Driving speeds were set to the posted speed limits, while walking used the pedestrian-accessible street network. PT and driving costs were estimated from separate sources. Based on VISTA 2012–2020, PT users in Victoria made an average of two trips per day; therefore, a daily fare of AUD 8 was represented as AUD 4 per trip. Driving costs were estimated using ATAP⁸ guidelines, Australian Institute of Petroleum (AIP)⁹ fuel consumption values, and average fuel prices in Victoria.

Cycling routes were derived using a behaviourally informed impedance model that accounted for Level of Traffic Stress (LTS) (Jafari et al., 2025). Link-level impedance was computed as segment length multiplied by a LTS weight, with additional penalties applied at unsignalised or high-stress intersections. Routes were optimised for minimum impedance subject to a 15% detour tolerance relative to the shortest path, allowing cyclists to trade modest increases in distance for improved perceived safety. The utility function for individual i choosing mode j is:

$$U_{ij} = ASC_j + \beta_{TT,j} \cdot T_{ij} + \beta_{money} \cdot C_{ij} + \beta_{LTS} \cdot P_{LTS,ij} \quad (12)$$

where ASC_j denotes the ASCs, T_{ij} is travel time, C_{ij} is monetary cost, and $P_{LTS,ij}$ represents the proportion of the cycling route exposed to high-stress road segments. Parameters $\beta_{TT,j}$, β_{money} , and β_{LTS} were estimated from VISTA.

To capture heterogeneity in travel behaviour, the estimation sample was segmented into six population clusters based on main activity and gender. Separate parameter sets were estimated for each cluster, enabling systematic differences in mode preferences across demographic groups. Table 3 summarises the resulting clusters and sample sizes. The estimated cluster-specific parameters were subsequently applied to assign travel modes within the CoGB population.

The ASC recalibration was implemented as a post-estimation adjustment step using observed aggregate mode shares in CoGB as calibration targets. Only the ASCs were adjusted in this step; the estimated coefficients on travel time, monetary cost, and cycling stress were retained from the Greater Melbourne model. This separation allows the model to adapt to regional differences in baseline mode preference without re-estimating behavioural trade-offs from a small local sample. The resulting comparison between pre- and post-recalibration mode shares is reported in Figure 11.

⁸https://www.atap.gov.au/sites/default/files/pv2_road_parameter_values.pdf

⁹<https://aip.com.au/aip-annual-retail-price-data>

Table 3: Summary of user clusters by main activity, gender, and sample size

Cluster	Main Activity	Gender	Sample Size (N)
1	Worker	Male	19,269
2	Worker	Female	19,645
3	Student	Male	6,949
4	Student	Female	6,998
5	Other	Male	5,166
6	Other	Female	8,196

4 Results

4.1 Activity chain results

Figure 6 compares the start and end time distributions of activities assigned to the activity-based demand with those observed in the VISTA 2012–2023 dataset. Figures 6a and b show the distributions for all activities, while Figures 6c and d focus only on non-home activities. Overall, the activity-based demand schedules closely reproduce the temporal patterns observed in VISTA, including the concentration of activity starts in the morning and peaks in activity endings during the afternoon and evening.

When all activities are considered, the model captures the broader daily rhythm of activity participation, including the expected dominance of home activities at the beginning and end of the day. After excluding home activities, the temporal distributions of non-home activity start and end times remain broadly consistent with the observed survey patterns. Minor deviations across some time bins are primarily attributable to limited observations in the VISTA dataset and the stochastic nature of the activity-generation process, rather than to systematic bias. Overall, these results indicate that the activity generation framework provides a behaviourally plausible representation of daily activity timing.

4.2 Location assignment

This section evaluates the location assignment results for both mandatory (work and education) and secondary activities. Validation is conducted using Census JTW and VISTA survey data, assessing the model’s ability to reproduce observed commuting distances, and regional origin–destination (OD) patterns.

Figure 7 compares the observed trip distance distribution derived from VISTA with the corresponding distribution generated in the activity-based demand. Assigned trips broadly match the observed distributions from VISTA, indicating that the integrated location assignment framework preserves aggregate travel distance patterns. This comparison should be interpreted as a plausibility check rather than a fully independent validation, because observed trip-length distributions also inform parts of the location assignment procedure. A slightly lower share of short trips and a higher share of longer trips are observed in the activity-based demand, reflecting the model’s explicit accounting for cross-boundary commuting outside the study region, which is underrepresented in the VISTA survey. As a result, the inclusion of external trips shifts the distribution modestly toward longer travel distances while preserving the overall pattern.

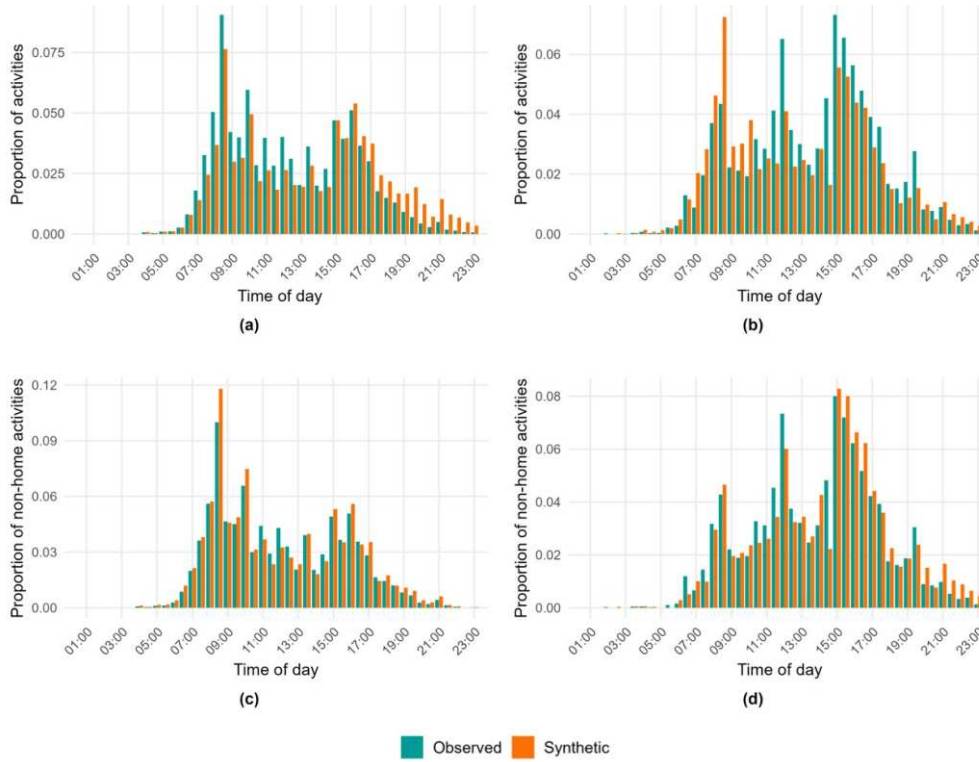


Figure 6: Comparison of activity timing distributions between the observed VISTA data and the activity-based demand: (a) start time of all activities, (b) end time of all activities, (c) start time of non-home activities, and (d) end time of non-home activities.

4.2.1 Mandatory location assignment results

Work location assignment was first validated against ABS JTW data by comparing commuting distance distributions across SA3 OD pairs. As reported in Table 4, the activity-based demand closely reproduces observed distance patterns, with mean absolute deviations remaining below one percentage point, with larger deviations concentrated in the shortest-distance bins. This confirms that the assignment procedure is internally consistent with the JTW-based calibration targets and reproduces the intended commuting structure. However, because SA3 flows and distance distributions derived from JTW data are used directly within the assignment process, this comparison should be interpreted as a calibration and internal consistency check rather than a fully independent validation.

For this reason, the comparison with VISTA and other non-targeted outcomes is particularly important, as it provides a more independent indication of whether the resulting workplace assignments also generate behaviourally plausible travel patterns beyond the calibration dimensions imposed by the model.

Beyond distance-based validation, spatial consistency was evaluated at the SA3 level by comparing home–destination shares for mandatory activities (education and work) against VISTA observations, with absolute differences shown in Figure 8. For both activity types, the VISTA data show strong intra-SA3 concentration, particularly for SA3 20201. In contrast, the activity-based demand allocates a slightly larger share of trips to neighbouring SA3s, resulting in

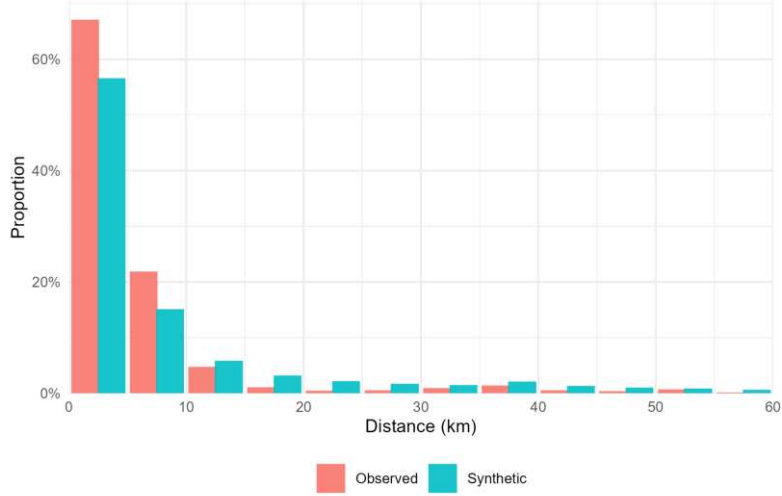


Figure 7: Observed and simulated distance histograms for trips after activity location assignments

Table 4: Percentage-point differences between observed and generated work trip distance distributions by SA3 OD pair.

SA3 _{home}	SA3 _{work}	Distance bins (km)																			
		4	8	12	16	20	24	28	32	36	40	44	48	52	56	60	64	68	72	76	80
20201	20201	2.68	2.36	1.31	0.56	0.06	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
20201	20202	0.00	0.06	0.05	0.09	0.03	0.04	0.03	0.11	0.12	0.10	0.13	0.11	0.01	0.01	0.01	0.03	0.04	0.02	0.01	0.06
20201	20203	0.00	0.08	0.00	0.01	0.04	0.03	0.06	0.05	0.02	0.01	0.03	0.03	0.02	0.02	0.00	0.02	0.02	0.02	0.06	0.00
20202	20201	0.03	0.10	0.53	0.77	0.75	0.46	0.28	0.11	0.48	0.61	0.53	0.31	0.22	0.12	0.22	0.44	0.75	0.33	0.15	0.48
20202	20202	0.34	0.15	0.37	0.91	0.37	0.28	0.33	0.19	0.37	0.22	0.21	0.14	0.30	0.24	0.24	0.47	0.18	0.13	0.14	0.03
20202	20203	0.00	0.01	0.00	0.01	0.01	0.02	0.01	0.01	0.01	0.02	0.00	0.01	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.00
20203	20201	0.00	0.05	0.14	0.26	0.12	0.12	0.03	0.04	0.15	0.22	0.06	0.08	0.06	0.02	0.02	0.00	0.00	0.08	0.23	0.02
20203	20202	0.00	0.01	0.00	0.01	0.01	0.02	0.01	0.01	0.02	0.01	0.01	0.00	0.04	0.01	0.01	0.01	0.01	0.00	0.01	0.00
20203	20203	0.16	0.08	0.04	0.00	0.01	0.15	0.03	0.05	0.04	0.07	0.01	0.12	0.02	0.01	0.06	0.02	0.01	0.01	0.03	0.02

Note: Distance-bin labels indicate the upper bound of each bin. For example, 4 represents 0–4 km, 8 represents 4–8 km, and so on.

differences along the diagonal and for some inter-SA3 flows.

For education trips, Figure 8 shows notable differences for some SA3 pairs, with the activity-based demand allocating a modest share of trips to neighbouring SA3s, while VISTA reports a much stronger concentration of intra-SA3 travel. A similar pattern is observed for work trips, with survey data suggesting stronger internal retention than the activity-based demand.

The activity-based demand was calibrated to reproduce Census JTW commuting flows and distance distributions, achieving close agreement across regional OD pairs as shown in Table 4. Since these are calibration targets, agreement on these dimensions is expected and should not be interpreted as independent proof of predictive validity. The comparison with VISTA is useful as a complementary check, although it is also subject to limitations arising from small effective sample sizes in regional survey data, which may overstate intra-SA3 retention and under-represent less common inter-SA3 movements.

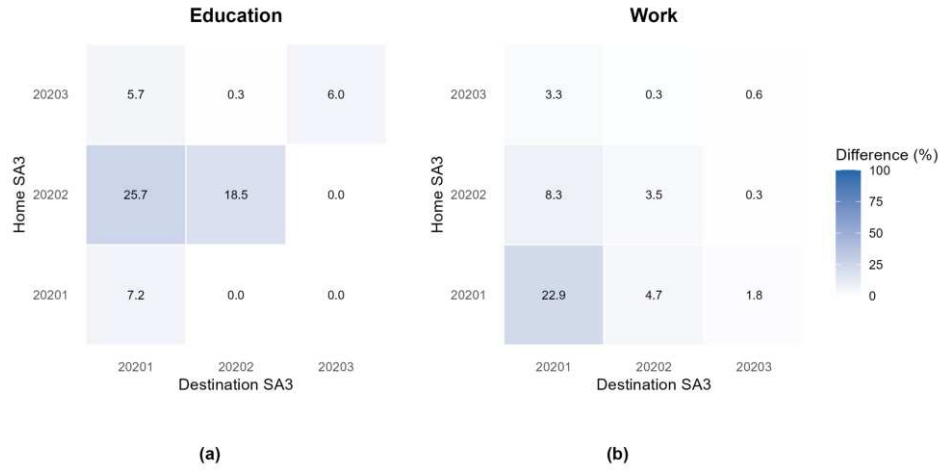


Figure 8: SA3-level absolute percentage-point differences in home–destination shares between the activity-based demand and VISTA for mandatory activities.

4.2.2 Secondary location assignment results

Figure 9 presents SA3-level absolute percentage differences in home–destination shares between the activity-based demand and VISTA survey data for secondary activities, including shopping, pickup/dropoff, social/recreational, and other purposes.

Across all secondary activity types, the largest relative differences are concentrated in trips originating in SA3 code 20201 and SA3 code 20202, particularly for shopping, pickup/dropoff, and social/recreational activities. In contrast, differences for many other SA3 origin–destination pairs remain comparatively small. Overall, the figure indicates that discrepancies are concentrated in a limited number of home–destination combinations rather than being widespread across all secondary travel patterns.

Several of the larger differences occur for inter-SA3 movements. This pattern mirrors that observed for mandatory activities and reflects the fact that secondary activity locations are conditioned on previously assigned work and education anchors within the gravity-based allocation framework.

Notably, these discrepancies are concentrated in the same SA3 regions identified during mandatory activity validation. This is partly expected because secondary locations are conditioned on previously assigned work and education anchors, so differences in the spatial distribution of mandatory destinations can propagate into subsequent non-mandatory travel. At the same time, the comparison is undertaken at a relatively coarse SA3 level and against a regional survey sample with limited effective observations for some inter-SA3 movements. The observed differences, therefore, likely reflect a combination of survey sparsity, coarse spatial aggregation, and the anchor-based propagation of mandatory location patterns, rather than a systematic failure of the secondary location assignment procedure.

Figure 10 illustrates the spatial distribution of activity locations across CoGB at 12:00, representing a midday snapshot of the activity-based demand. At this time of day, home activities are comparatively fewer than during morning and evening periods, while work and study locations dominate the central urban area. A clear concentration of employment and education activities is visible within the CoGB CBD and surrounding mixed-use zones.

Secondary activities such as shopping, pickup/dropoff, and social trips appear distributed around commercial

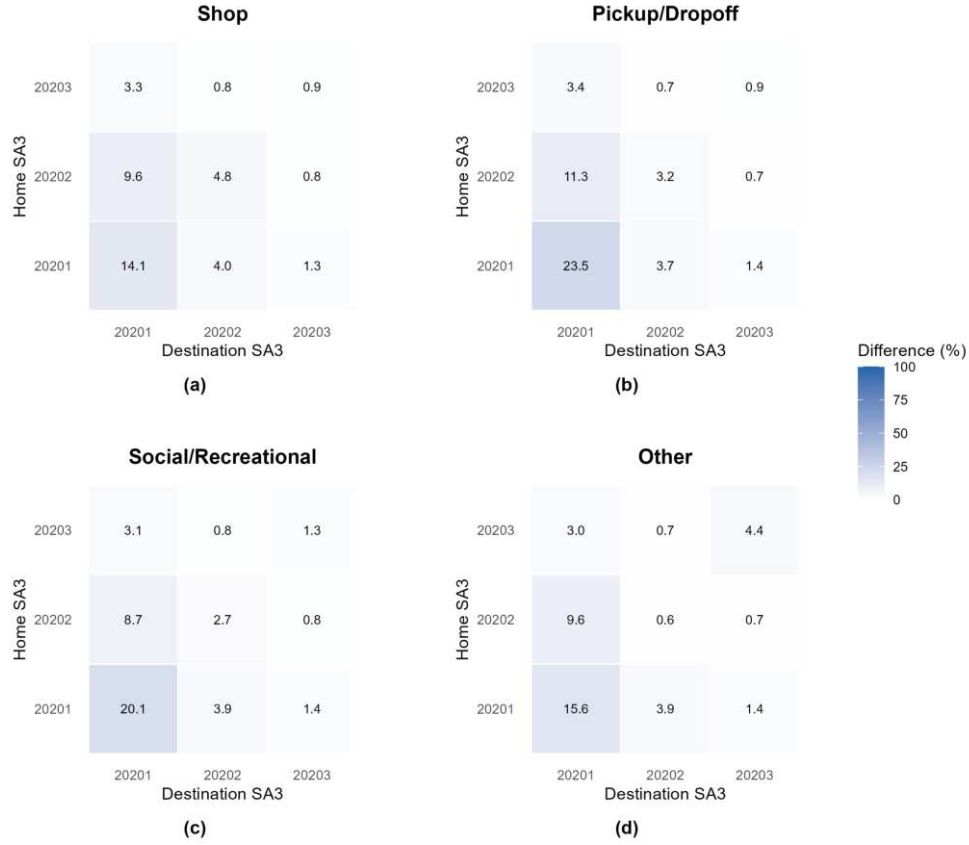


Figure 9: SA3-level absolute percentage-point differences in home-destination shares between the activity-based demand and VISTA for secondary activities.

centres and along major transport corridors. Residential locations remain more spatially dispersed across suburban areas, reflecting the underlying population distribution. Overall, the spatial patterns illustrate the expected daytime shift from dispersed residential origins to more concentrated activity centres, indicating that the location assignment framework generates coherent and behaviourally plausible activity patterns across the region.

4.3 Mode choice model

Table 5 reports the estimated MNL models for the six user clusters defined in Table 3. Model fit statistics indicate satisfactory explanatory power (R^2 between 0.53 and 0.65), supporting the use of these models for mode assignment in the activity-based demand. The sample sizes reported in the table refer to the unweighted number of observed survey trips used for estimation, whereas the MNL models were estimated using survey expansion weights, so the fitted parameters reflect the weighted travel population. Across almost all clusters, travel time coefficients are negative and statistically significant, with only a few insignificant estimates, confirming the strong disutility generally associated with longer travel times. Sensitivity to travel time varies across clusters, with the student group (Cluster 3) exhibiting comparatively lower time disutility for driving and PT, reflecting higher temporal flexibility and potentially lower opportunity costs of travel time relative to workers and other groups.

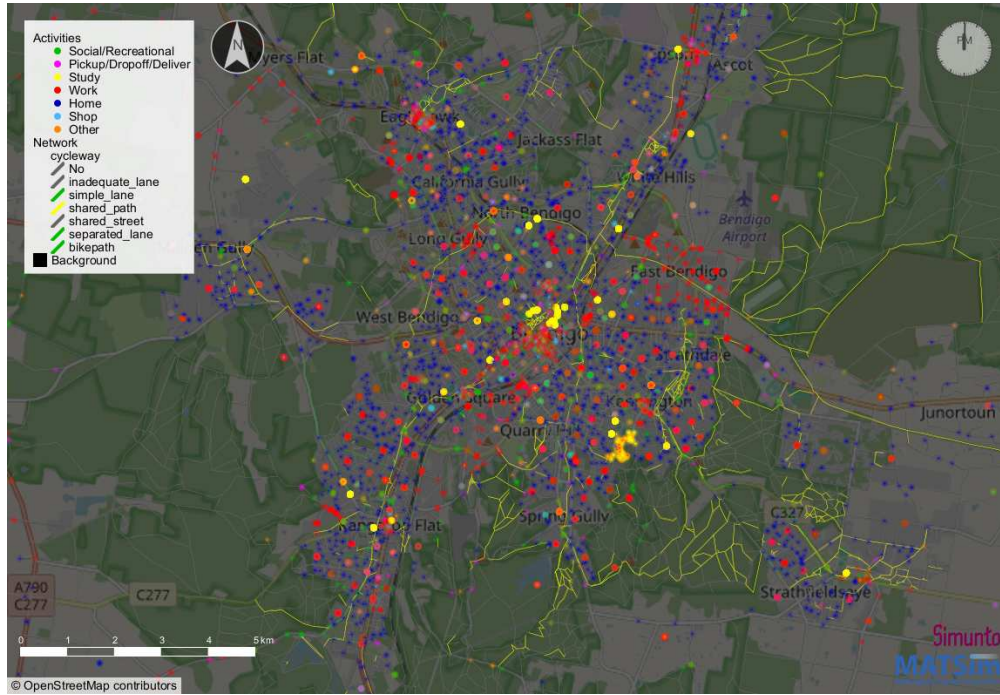


Figure 10: Spatial distribution of activity locations across CoGB at 12:00 PM

Monetary cost coefficients are negative and statistically significant across all clusters, indicating systematic sensitivity to travel costs, with comparatively higher cost aversion observed among non-working females (Cluster 6) and lower sensitivity among student groups (Clusters 3 and 4). ASCs reveal a strong baseline preference for walking and systematically negative constants for cycling, highlighting persistent barriers to cycling adoption in CoGB. These effects likely reflect unobserved factors such as safety perceptions, habits, and comfort (Tiwari et al., 2026).

The coefficient associated with exposure to high-stress cycling infrastructure (β_{LTS}) exhibits clear heterogeneity across clusters. It is negative and statistically significant for Clusters 1 (working males), 2 (working females), and 6 (non-working females), indicating a strong aversion to stressful cycling environments among workers and non-working females, whereas it remains insignificant for student clusters (Clusters 3 and 4) and non-working males (Cluster 5), suggesting lower sensitivity to traffic stress within these groups.

4.3.1 Mode share validation and ASC calibration

The estimated parameters from Table 5 were applied to assign modes to each trip. Using the transferred Melbourne ASCs without adjustment leads to systematic differences between simulated and observed mode shares in CoGB. The baseline simulation reproduces only 77.8% car mode share compared with the observed 83.8%, while overestimating PT (5.9% vs. 3.3%) and walking (15.3% vs. 11.6%) mode shares. These discrepancies reflect structural differences in baseline modal attractiveness between metropolitan and regional contexts, indicating that direct transfer of ASCs leads to biased mode shares.

To correct this baseline mismatch while preserving behavioural sensitivities, ASCs for PT and walking were iteratively adjusted across all clusters until aggregate simulated shares aligned with observed CoGB values, with all

Table 5: Estimated MNL mode choice parameters across six user clusters

Coefficient	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 6
Cluster description	Male Worker	Female Worker	Male Student	Female Student	Male Others	Female Others
Sample Size (N)	19269	19645	6949	6998	5166	8196
ASC _{PT}	0.172 (0.189)	0.341 (0.218)	-1.284*** (0.084)	-1.838*** (0.086)	-0.620** (0.202)	-0.071 (0.181)
ASC _{cycling}	-2.502*** (0.126)	-2.854*** (0.157)	-2.293*** (0.061)	-3.110*** (0.089)	-3.807*** (0.116)	-4.546*** (0.153)
ASC _{walking}	1.146*** (0.065)	1.078*** (0.063)	1.113*** (0.028)	1.251*** (0.030)	1.102*** (0.037)	1.487*** (0.032)
$\beta_{\text{trav, driving}}$	-0.044*** (0.005)	-0.051*** (0.005)	-0.006** (0.008)	-0.037*** (0.003)	-0.039*** (0.005)	-0.019*** (0.004)
$\beta_{\text{trav, PT}}$	-0.042*** (0.002)	-0.044*** (0.002)	0.000 (0.002)	-0.009*** (0.001)	-0.033*** (0.002)	-0.031*** (0.002)
$\beta_{\text{trav, cycling}}$	-0.060*** (0.003)	-0.077*** (0.005)	-0.091*** (0.003)	-0.101*** (0.005)	-0.051*** (0.003)	-0.066*** (0.006)
$\beta_{\text{trav, walking}}$	-0.081*** (0.002)	-0.085*** (0.002)	-0.083*** (0.001)	-0.095*** (0.001)	-0.087*** (0.001)	-0.108*** (0.001)
β_{money}	-0.399*** (0.041)	-0.460*** (0.047)	-0.382*** (0.020)	-0.460*** (0.047)	-0.403*** (0.044)	-0.557*** (0.040)
β_{LTS}	-0.892*** (0.233)	-1.445*** (0.312)	-0.099 (0.099)	0.036 (0.138)	-0.313 (0.201)	-0.553* (0.270)
Model fit statistics						
Log-likelihood (LL)	-8209.47	-7997.91	-36517.21	-33846.42	-18254.22	-25031.85
R^2	0.5709	0.5792	0.5345	0.5705	0.6005	0.6502
AIC	16436.93	16013.82	73052.41	67710.84	36526.45	50081.69

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

Note: Driving is the reference alternative. Standard errors are reported in parentheses.

other coefficients fixed. This procedure preserved the relative effects of travel time, cost, and infrastructure attributes while accounting for systematic differences in baseline mode preferences.

The recalibrated constants are reported in Table 6. After calibration, simulated mode shares closely reproduce the observed distribution in CoGB, substantially reducing the overrepresentation of PT and walking and correcting the underprediction of car use. Figure 11 illustrates the comparison between observed and simulated mode shares before and after ASC calibration. Table 7 shows that ASC recalibration substantially improved the match between simulated and observed mode shares in CoGB, reducing the mean absolute error from 3.15 to 0.11 percentage points.

Table 6: Original and recalibrated ASCs for PT and walking

Cluster	ASC _{PT} (Original)	ASC _{PT} (Recalibrated)	ASC _{walk} (Original)	ASC _{walk} (Recalibrated)
Cluster 1	0.172	-0.193	1.146	0.067
Cluster 2	0.341	-0.181	1.078	0.560
Cluster 3	-1.284	-2.994	1.113	0.518
Cluster 4	-1.838	-3.098	1.251	0.410
Cluster 5	-0.620	-1.850	1.102	0.611
Cluster 6	-0.071	-1.591	1.487	0.927

4.4 External trip integration results

Table 8 summarises the percentage distribution of work trips between home and work SA3 regions used in the demand generation process. In the generated demand, approximately 12% of work trips originating within the CoGB SA3 regions have workplaces located outside the study area, highlighting strong labour-market interactions between

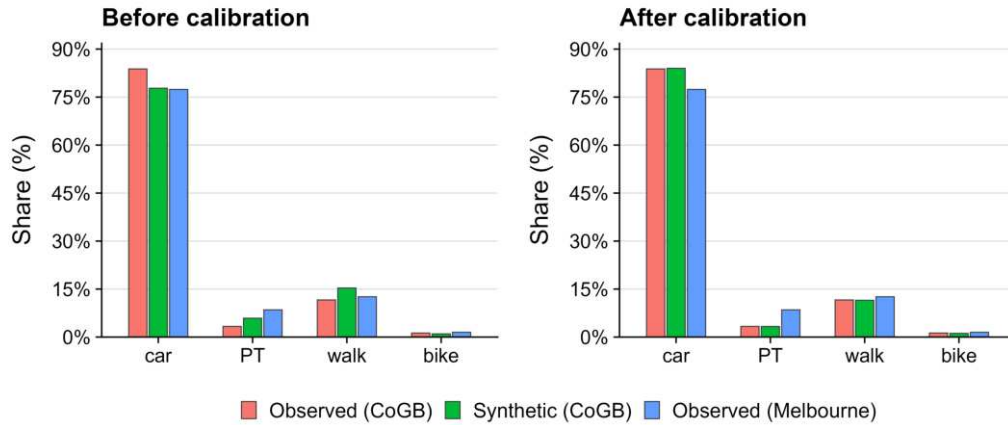


Figure 11: Mode share validation for CoGB. Baseline results using Melbourne-calibrated ASCs (left) and after ASC recalibration (right).

Table 7: Mode share comparison before and after ASC recalibration (CoGB).

Mode	Observed	Generated Before Calibration		Generated After Calibration	
		Share	Abs. Error	Share	Abs. Error
Car	83.80	77.77	6.03	84.00	0.20
PT	3.34	5.89	2.55	3.32	0.02
Walk	11.60	15.34	3.74	11.50	0.10
Bike	1.26	1.00	0.26	1.16	0.10
MAE	–	–	3.15	–	0.11

CoGB and surrounding regions.

The table also shows that around 1% of trips represent workers commuting into CoGB from outside the study region. Although smaller than outbound commuting, this inflow still represents a meaningful component of regional travel demand. Taken together, about 13% of total work trips involve cross-boundary commuting, representing more than 8,000 daily commuting trips in the activity-based demand. Explicitly accounting for these flows is therefore important for preserving the regional structure of commuting patterns and ensuring that the activity-based demand captures travel demand on connections between CoGB and neighbouring regions.

These shares should be interpreted as a modelled representation of the JTW-based commuting structure rather than as an independent empirical finding. Their value lies in showing how cross-boundary commuting is carried through into the synthetic activity-based demand and subsequent activity chains.

Table 8: Percentage distribution of work trips between home SA3 regions and work locations

Home SA3	Work SA3			
	20201	20202	20203	Other regions
20201	52.3	4.1	1.5	3.0
20202	7.3	15.0	0.3	8.2
20203	2.9	0.2	3.3	0.8
Other regions	0.3	0.8	0.1	0.0

5 Discussion

This study developed a framework for generating activity-based transport demand suitable for regional ABM where detailed behavioural data are often limited. By combining demographic synthesis, probabilistic activity-chain generation, hierarchical location assignment, and transferred mode choice modelling, the framework produces simulation-ready activity-based demand for CoGB while explicitly accounting for regional boundary effects. More broadly, the study contributes to a growing literature on open and reproducible activity-based demand generation for ABM, which has so far been dominated by applications in large metropolitan regions such as Berlin, Paris, São Paulo, and Melbourne (Ziemke et al., 2019; Hörl and Balac, 2021; Sallard et al., 2021; Both et al., 2021). Compared with these studies, the present contribution is less focused on large-scale metropolitan completeness and more focused on a regional modelling problem in which cross-boundary travel and sparse local data are central constraints.

A central challenge in regional ABM is balancing spatial coverage with local realism. Regional cities are strongly connected to surrounding areas and nearby major cities through commuting, education, and access to specialised services, yet available behavioural data are often sparse and geographically incomplete. Models limited to small study areas risk truncating external trips and artificially localising travel patterns, while broader models often sacrifice neighbourhood-level detail to remain computationally tractable (Baqueri et al., 2019). The framework proposed in this study addresses this trade-off by explicitly incorporating cross-boundary commuting from Census JTW data while retaining detailed local assignments for homes, workplaces, educational locations, and secondary activities. In the CoGB case, approximately 13% of work trips involved cross-boundary commuting, indicating that external travel is not a marginal feature but a structurally important component of regional demand.

The results also highlight the value of assigning activities hierarchically rather than independently. As in other recent activity-based demand frameworks, daily activity and travel decisions are interdependent, with earlier choices shaping the feasible set of later choices (Both et al., 2021; Bowman and Ben-Akiva, 2001). Recent work has also shown that discretionary destination choice can depend not only on destination attractiveness and travel impedance from the current location, but also on the locations of activities immediately before and after in the schedule (Moeckel et al., 2024). In the proposed framework, work and education locations act as primary spatial anchors of the daily schedule and strongly shape the feasible geography of subsequent discretionary travel. By assigning these mandatory activities first, then conditioning secondary destination choice on both preceding and subsequent anchors, the framework generates more coherent daily trajectories than approaches that allocate non-mandatory destinations independently of the surrounding chain. This is particularly important in regional settings, where the limited number and spatial concentration of major destinations can otherwise produce implausible detours or overly dispersed activity patterns.

The work location assignment results also demonstrate the utility of combining aggregate commuting constraints with local stochastic assignment. Existing activity-based demand studies have shown the value of using open or aggregate data to construct spatially detailed demand in the absence of workplace microdata (Ziemke et al., 2019; Hörl and Balac, 2021; Sallard et al., 2021). The present study extends that logic by jointly imposing SA3 commuting flows, conditional distance distributions, and adaptive local attractiveness balancing. This produces workplace patterns that are consistent with observed regional labour market structure while retaining stochastic variation at finer spatial scales. At the same time, it is important to recognise that agreement with JTW-based commuting targets should be interpreted primarily as evidence of internal consistency with the assignment design, rather than as a fully independent validation, because these targets are used directly within the work location assignment procedure. The more informative contribution is therefore not simply that the model matches these targets, but that it does so while still generating plausible activity chains, non-mandatory travel patterns, and aggregate mode shares across the full demand generation process.

The education and secondary activity results illustrate both the strengths and the limitations of the proposed approach. Capacity-constrained assignment produced plausible education patterns, while the two-sided gravity-based method generated secondary locations consistent with observed trip-length distributions and the spatial logic of daily schedules. At the same time, some differences remained between the activity-based demand and VISTA, particularly for inter-SA3 movements. These differences likely reflect several factors: limited regional survey sample sizes, the coarse spatial scale of validation, and the fact that secondary destinations inherit the spatial structure of previously assigned mandatory anchors. The discrepancies should therefore not be interpreted solely as model error, but rather as the outcome of comparing an activity-based demand calibrated partly to Census commuting data against a smaller survey sample that may under-represent some regional movements.

The transferred mode-choice model is another pragmatic component of the framework. Rather than estimating a local model from a small CoGB sample, behavioural sensitivities were transferred from Greater Melbourne, and only the ASCs were recalibrated to local mode shares. This approach assumes that the underlying behavioural responses to key mode attributes, including time, cost, and cycling stress, are broadly transferable, while differences in observed mode shares arise mainly through differences in urban form, network structure, and resulting level of service. This is a stronger assumption than full local estimation, but it is more defensible than transferring an entire demand model without recalibration. It is also broadly consistent with findings from transferability research in activity-based modelling, which suggest that model structures can often be transferred across contexts, but that recalibration is usually required when land use patterns, mode availability, and local preferences differ substantially (Galbraith and Hensher, 1982; San Santoso and Tsunokawa, 2005). In this study, the mode-share validation suggests that such a compromise is workable for data-scarce regional settings, although future work should test this more directly using larger local stated- or revealed-preference datasets, where available.

Several limitations should be acknowledged. First, the framework relies on a combination of datasets collected in different years, including Census JTW for 2016, VITM2 for 2018, and VISTA in 2012 and 2023, and therefore assumes that the main behavioural and spatial structures are sufficiently stable across these periods for activity-based demand. Second, activity schedules are generated before location and mode assignment, meaning that the generated timings represent desired daily activity patterns rather than fully time-adjusted realised schedules. Travel times are not used to reschedule activities during demand generation. In a downstream agent-based simulation, network travel times and replanning can further adjust realised behaviour, but future work could integrate explicit schedule-feasibility checks after location and mode assignment. Third, primary school assignment is represented through a nearest-feasible rule, which does not explicitly capture school catchments, private school attendance, or parental choice. Fourth, the secondary activity model uses balancing parameters and equal anchor weights chosen

for robustness and transparency rather than formal behavioural estimation. It also uses a provisional mode during location assignment to define reachability constraints, while final mode assignment is undertaken later through the MNL model. This may introduce some inconsistency between the spatial-feasibility mode and the final assigned mode, although the purpose of the provisional mode is only to avoid implausible secondary destination distances. Fifth, validation of non-mandatory travel remains constrained by the small effective sample sizes available in regional household travel survey data. Sixth, mode choice parameters are transferred from a metropolitan VISTA sample of Greater Melbourne, assuming that behavioural sensitivities are transferable to the regional context; while ASCs are recalibrated, potential differences in behavioural responses are not explicitly captured.

Despite these limitations, the framework offers a practical and transferable contribution to regional activity-based transport modelling. It demonstrates how publicly available census, survey, land use, and network data can be combined to construct behaviourally plausible activity-based demand for regional cities where conventional metropolitan modelling approaches are difficult to apply directly. More broadly, the study shows that explicit representation of cross-boundary flows, hierarchical anchoring of activities, and pragmatic behavioural transfer can substantially improve the realism of regional demand generation for ABM.

Future research could extend the framework in several directions. These include incorporating richer school choice mechanisms, testing alternative specifications for secondary activity weighting and anchor influence, improving validation of non-mandatory travel with larger behavioural datasets, and examining the transferability of the framework to other regional cities with different spatial structures and travel patterns. Explicit testing of the transferability of mode-choice parameters between Greater Melbourne and CoGB, or the estimation of region-specific models when sufficient local data are available, would further strengthen behavioural realism. Additional work could also integrate joint household decision-making and more detailed representations of destination choice and activity participation. Overall, the framework provides a practical basis for developing regional ABM capable of supporting transport planning and policy analysis in data-constrained contexts.

6 Conclusion

This paper presented a regional activity-based transport demand generation framework for use in agent-based transport simulation models, using CoGB as a case study. The framework combines demographic synthesis, probabilistic activity-chain generation, hierarchical assignment of mandatory and non-mandatory activity locations, explicit representation of cross-boundary commuting, and a transferred mode choice model with local ASC recalibration. Together, these components produce an activity-based demand that reproduces key calibration benchmarks and aligns with several observed patterns, including population totals, activity timing patterns, commuting distances, regional commuting flows, and aggregate mode shares.

The main methodological contribution lies in integrating coarse regional commuting information with fine-grained local activity assignment. Work and education locations are assigned first to establish spatial anchors of the daily schedule, after which secondary activities are allocated using a two-sided gravity-based procedure conditioned on both preceding and subsequent anchors. This structure improves the spatial coherence of daily trajectories and is particularly useful in regional contexts, where travel is shaped by both concentrated local destinations and strong cross-boundary interactions.

The CoGB application demonstrates that realistic synthetic demand can be generated in a regional city using largely publicly available data, without requiring either a large local travel survey or a full statewide microsimulation. The results also underscore the importance of explicitly accounting for external travel, as cross-boundary commuting accounts for a substantial share of work trips in the study region.

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Disclosure of interest

The authors declare that they have no competing interests.

Data availability statement

Data are available on request from the authors.

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